MIL-C-81842A(AS)
1 September 1976
Superseding
MIL-C-81842(AS)
5 July 1972
Amendment-2
10 October 1972

MILITARY SPECIFICATION

CONNECTOR ASSEMBLIES FOR BOMB RACK ELECTRIC FUZING PROVISIONS

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE AND CLASSIFICATION

1.1 **Scope.** This specification establishes the requirements for the manufacture and inspection of connector assemblies. The test limits in this specification define minimum acceptable capabilities.

1.2 **Classification.** Connector assemblies covered by this specification shall be of the following types:

Type I	DL67A33F100, Connector Assembly
Type II	DL67A34F100, Connector Assembly
Type III	DL67A35D100, Connector Assembly
Type IV	DL67A37D110, Connector Assembly

2. APPLICABLE DOCUMENTS

2.1 **Issues of documents.** The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to:

ENGINEERING SPECIFICATIONS AND STANDARDS DEPARTMENT (CODE 93) NAVAL AIR ENGINEERING CENTER, LAKEHURST, N.J. 08733

by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1095

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SPECIFICATIONS

Military

MIL-R-81939	Rack, Bomb Ejector, Aircraft, BRU 10, BRU 11 and BRU 19 Series
MIL-R-22622	Rack, Bomb Ejector, Aircraft, Aero 7 Series
MIL-R-22623	Rack, Bomb Ejector, Aircraft, Aero 20 Series
AS-2564	Bomb Rack Assembly, Multiple Ejector A/A37B-6
AS-2565	Bomb Rack Assembly, Triple Ejector A/A37B-5

STANDARDS

Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation of

Drawings

DL 1011AS100	Data List, Bomb Rack Electric Fuzing Vibration Fixture

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Order of precedence of specifications and standards.** Should a conflict exist between any requirement in this specification and any requirement in the applicable documents referenced herein, the requirements of this specification shall take precedence over the referenced (applicable) document.

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3. REQUIREMENTS

3.1 First article. The connector assemblies furnished under this specification shall be a product which has been inspected and passed the first article inspection specified herein (see 4.3).

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Interchangeability. All parts having the same part number shall be directly and completely interchangeable with each other with respect to installation and performance.

3.4 Identification of parts. All parts shall be marked as specified on the drawings listed in the data list for the connector assemblies being procured, or in accordance with MIL-STD-130, if not specified on the drawings. The item identification and part number requirements of MIL-STD-100 shall govern the part number and changes thereto.

3.5 Connector assembly.

3.5.1 Construction. The construction of the connector assemblies shall be in accordance with the drawings listed in the data list for the connector assembly being procured, and as specified herein (see 6.2.1f).

3.5.2 Performance. The connector assemblies shall satisfy all performance requirements when subjected to the following inspections in the order specified in Tables I and II.

a. Inspection, Connector Assembly (4.6.1)

b. Connector Pull-Out Force, 0 degrees (4.6.2.1)

c. Connector Pull-Out Force, 15 degrees (4.6.2.2)

d. Dielectric Withstanding Voltage (4.6.2.3)

e. Continuity Resistance (4.6.2.4)

f. Insulation Resistance (4.6.2.5)

g. High Temperature (4.6.3.1)

h. Low Temperature (4.6.3.2)

i. Humidity (4.6.3.3)

j. Sand (4.6.3.4)

k. Salt Fog (4.6.3.5)

l. Vibration (4.6.3.6)

m. Icing (4.6.3.7)

n. Life (4.6.4)

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TABLE I

FIRST ARTICLE INSPECTIONS - SCHEDULE AND SEQUENCE 1/

TESTS	TEST PARAGRAPH	SAMPLE		
		A	B	C
Connector Assembly Inspection	4.6.1	1	1	1
0° Connector Pull-Out Force	4.6.2.1	2	2	2
15° Connector Pull-Out Force	4.6.2.2	3	3	3
Dielectric Withstanding Voltage	4.6.2.3	4	4	4
Continuity Resistance	4.6.2.4	5	5	5
Insulation Resistance	4.6.2.5	6	6	6
High Temperature	4.6.3.1	7	X	X
Low Temperature	4.6.3.2	8	X	X
Humidity	4.6.3.3	X	7	X
Sand and Dust	4.6.3.4	9	X	X
Salt Fog	4.6.3.5	X	X	7
Vibration	4.6.3.6	10	8	8
Icing	4.6.3.7	11	9	9
Life	4.6.4	12	10	10

1/ Inspection sequence is denoted by the numbers in the sample columns. An X denotes inspection not required.

3.6 Workmanship. The workmanship displayed in fabrication and assembly of the connector assemblies shall be such as to assure, within design limitations, the ability of the connector assemblies to meet their performance requirements under all applicable environmental conditions specified herein. Unauthorized repair, heavy burrs or parts assembled by introduction of high stresses not prescribed in the drawings, are typical signs of inferior workmanship and shall be cause for rejection. The standards of workmanship exhibited in the approved first article sample, subject to any qualification stated in the government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship.

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TABLE II

QUALITY CONFORMANCE INSPECTIONS - SCHEDULE AND SEQUENCE 1/

TESTS	TEST PARAGRAPH	SAMPLE				
GROUP 1		All Samples				
Connector Assembly Inspection	4.6.1	1				
0° Connector Pull-Out Force	4.6.2.1	2				
15° Connector Pull-Out Force	4.6.2.2	3				
Dielectric Withstanding Voltage	4.6.2.3	4				
Continuity Resistance	4.6.2.4	5				
Insulation Resistance	4.6.2.5	6				
GROUP 2		A	B	C	D	E
Humidity	4.6.3.3	X	1	X	1	X
Vibration	4.6.3.6	1	X	1	X	1
Icing	4.6.3.7	2	X	2	X	2

1/ Inspection sequence is denoted by the numbers in the sample columns. An X denotes inspection not required.

4. QUALITY ASSURANCE PROVISIONS (see 6.4)

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection of the connector assemblies shall be classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

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4.3 First article inspection. First article inspection shall consist of all of the inspections listed in Table I. First article inspection report format shall be in accordance with MIL-STD-831 (see 6.3).

4.3.1 Sampling for first article inspection. First article connector assembly samples shall consist of the first three connector assemblies manufactured on contract. The connector assemblies shall be permanently labeled A, B and C for ease of identification. Failure of the connector assemblies to pass any of the first article inspections shall be cause for rejection (see 6.3).

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the following inspections:

a. Individual inspections (see 4.4.1).

b. Sampling inspections (see 4.4.2).

Quality conformance inspection report format shall be in accordance with MIL-STD-831 (see 6.3).

4.4.1 Individual inspections. Individual inspections are those inspections conducted on each connector assembly. Individual inspections for the connector assemblies are shown in Table II, Group 1. Failure to pass any of these inspections shall be cause for the rejection of the connector assembly.

4.4.2 Sampling inspections. A random sampling of the connector assemblies shall be selected by the procuring activity from each production lot in accordance with Table III and shall be subjected to the inspections listed in Table II, Group 1 and Group 2. Group 1 inspections shall be conducted prior to Group 2 inspections. Failure to pass any of these inspections shall be cause for rejection of the entire lot (see 6.3).

4.5 Inspection conditions. Unless otherwise specified, all inspections shall be performed under the following conditions:

a. Room temperature. Room ambient of 25 ± 10°C (77 ± 18°F) indicated.

b. Inspection temperature. All parts of the item shall be stabilized at the specified temperature ±2.8°C (±5°F) indicated prior to conducting any inspections. Unless otherwise specified, temperature stabilization will have been attained when the indicated temperature of the surface of the largest mass of the item does not change more than ±2.8°C (±5°F) per hour.

c. Humidity. Room ambient to 95 percent relative humidity. When humidity testing is required, the tolerance shall be ±5 percent relative humidity.

d. Connector assembly installation. When installation of the connector assembly in an applicable bomb rack is required, the bomb rack indicated in Table IV shall be utilized. The applicable bomb rack (and its ejection cartridges) shall be supplied by the procuring activity.

e. Electrical connectors and mating cables. When installed in the applicable bomb rack, the applicable mating cable (see Figures 1 through 4) shall be attached to the electrical connector unless otherwise specified. During salt fog, sand and humidity inspections, the end of the mating cables not attached to the connector assembly shall be sealed and moisture proof.

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TABLE III

SAMPLING TEST ITEMS

ITEM	PRODUCTION LOT SIZE	NUMBER OF SAMPLES
Connector Assemblies	100 or Less	3
	101 - 200	4
	201 - 300	5
	301 or More	1/

1/The procuring activity shall specify the number of samples, and the sequence of inspections each unit shall be subjected to for lots in excess of 300 units (see 6.2.1e).

TABLE IV

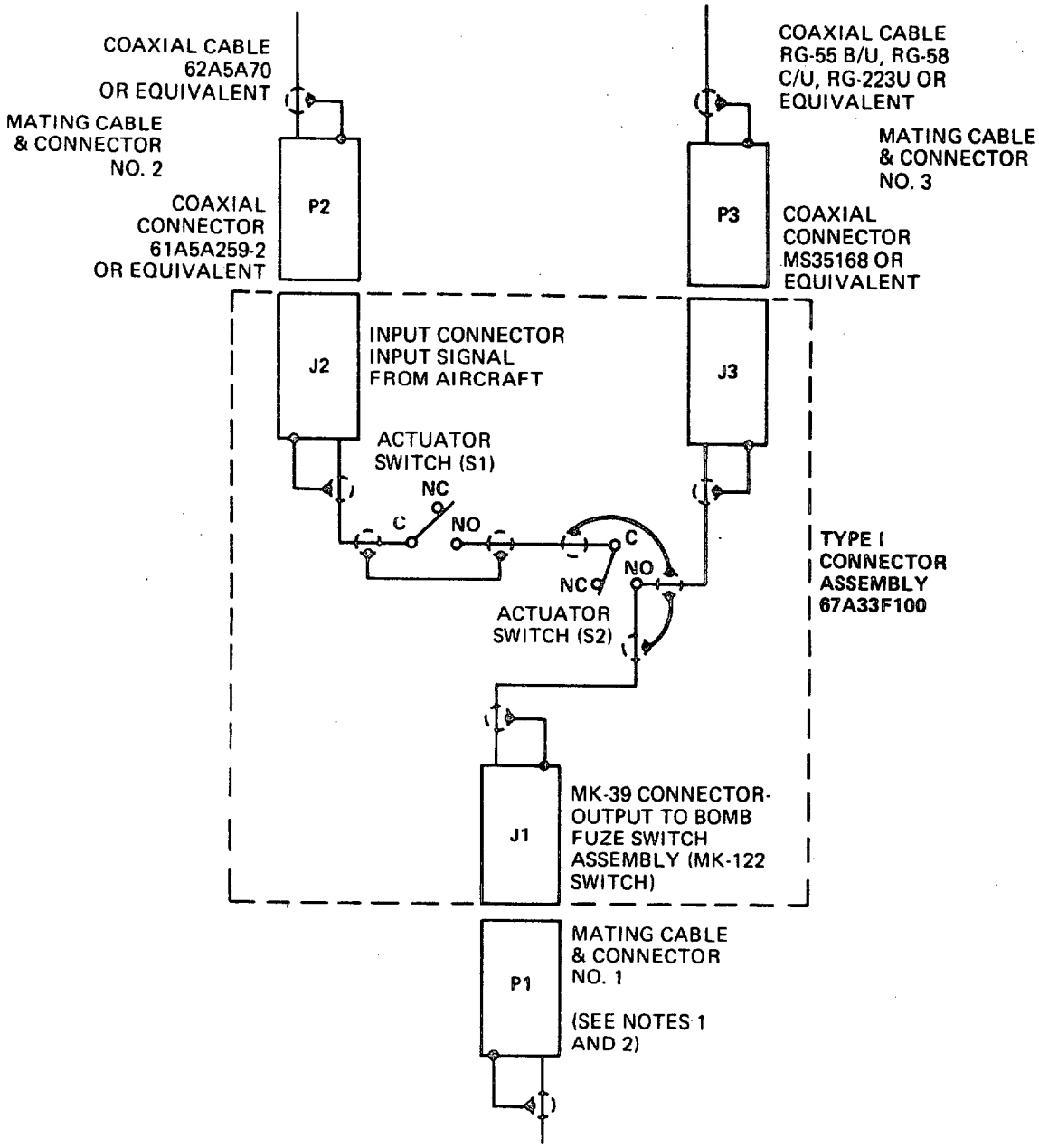
APPLICABLE BOMB RACKS

CONNECTOR ASSEMBLY	APPLICABLE PARENT BOMB RACK	MILITARY SPECIFICATION FOR APPLICABLE PARENT BOMB RACK
67A33F100	Aero 7A Series 1/ Aero 7B Series	MIL-R-22622
67A34F100	Aero 20 Series 1/	MIL-R-22623
67A35D100	BRU 10, 11 and 19 Series 1/	MIL-R-81939
67A37D110	A/A37B-5 Series 1/ 2/	AS-2565
	A/A37B-6 Series 1/ 2/	AS-2564

1/Swaybracing of stores required during connector assembly inspections shall be in accordance with the detailed specification referenced in this table for the parent bomb rack being used.

2/Used with MER/TER ejector units.

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NOTES:

(1) THE CONNECTOR OF A MK-122, MOD 0 SWITCH, P/N 2288036, AND THE COAXIAL CABLE ATTACHED (OR THE EQUIVALENT) COMPRISE THE NO. 1 MATING CABLE. THE CONNECTOR MAY BE OBTAINED FROM SHIPS PARTS CONTROL CENTER, MECHANICSBURG, PA.

(2) IT IS PERMISSIBLE TO MACHINE THE OUTER CONTOUR OF THE MK-122, MOD 0 SWITCH CONNECTOR TO FACILITATE INSERTION AND REMOVAL OF THE CONNECTOR FROM THE MK-39 RECEPTACLE OF THE CONNECTOR ASSEMBLY. THIS MATING CONNECTOR SHALL BE USED FOR ELECTRICAL TESTS ONLY.

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Figure 1. Mating Cables for Type I Connector Assembly

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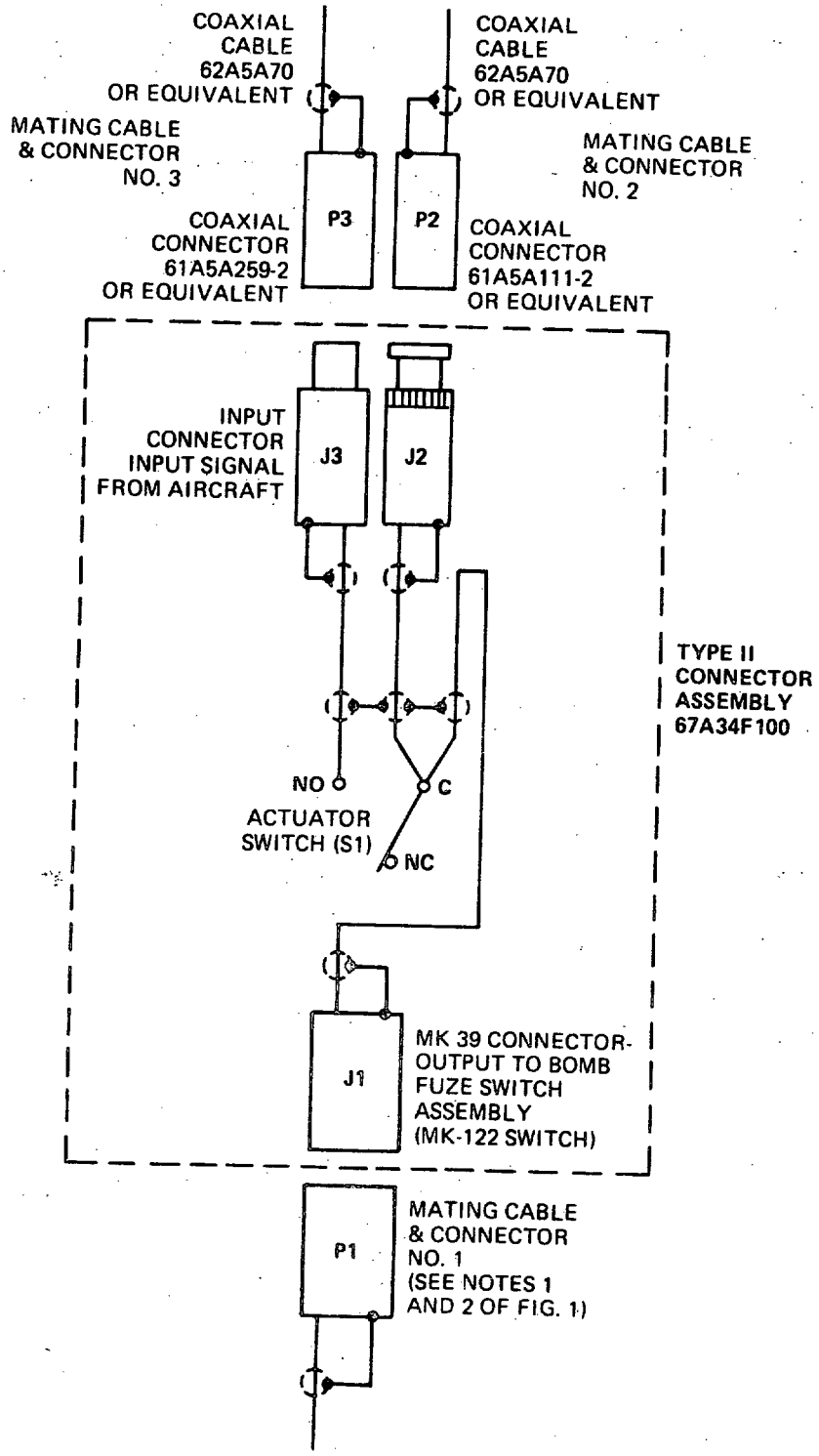


Figure 2. Mating Cables for Type II Connector Assembly

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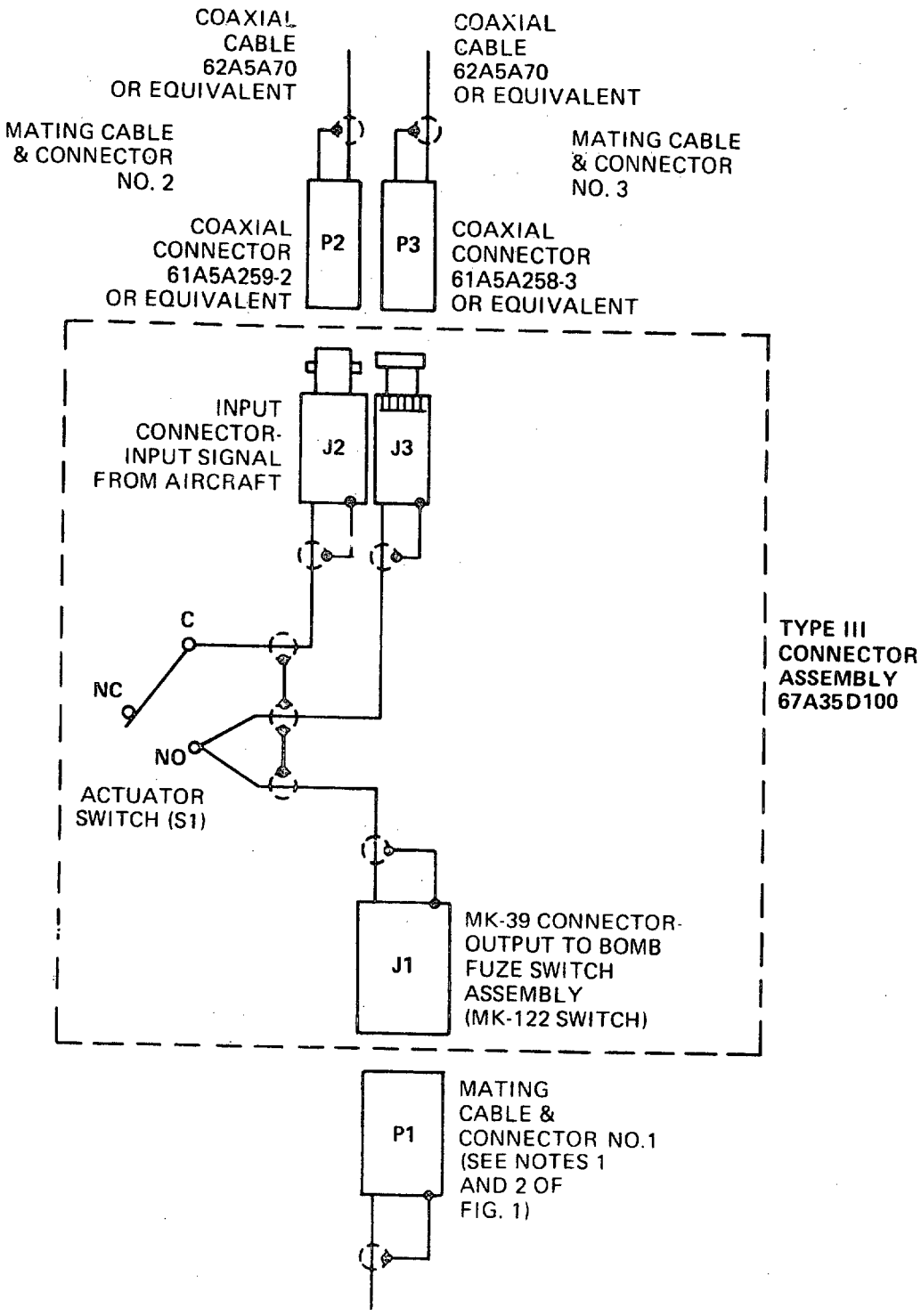


Figure 3. Mating Cables for Type III Connector Assembly

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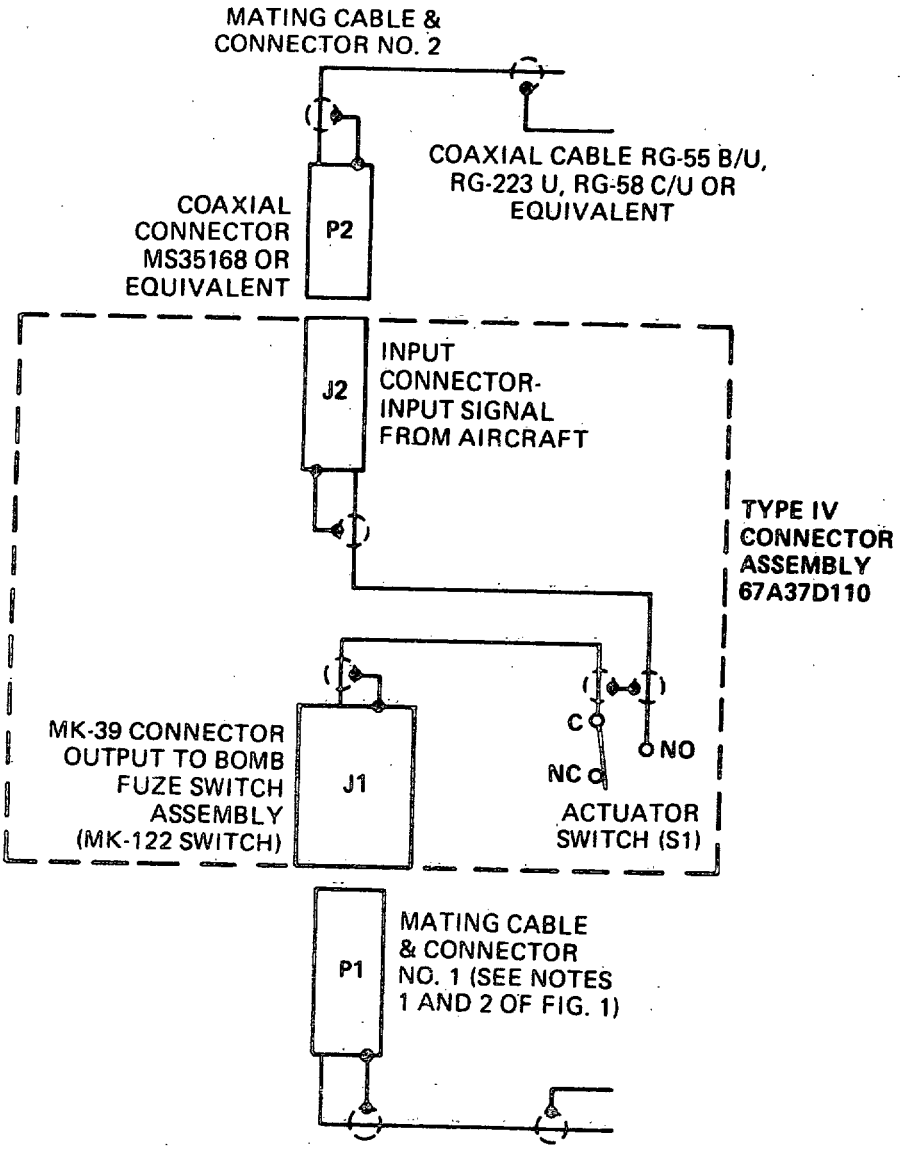


Figure 4. Mating Cables for Type IV Connector Assembly

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f. Test fixtures.

1. For the vibration inspection of 4.6.3.6, the connector assembly shall be mounted and adjusted as shown in DL 1011AS100 (see 2.1).

2. For the humidity inspection of 4.6.3.3 and the salt fog inspection of 4.6.3.5, the connector assembly shall be mounted in the test fixture shown in Figure 5 or, at the discretion of the procuring activity, in the applicable bomb rack of Table IV.

3. Pull-out force measurements (see 4.6.2.1 and 4.6.2.2), whenever required, shall be made with the pull test fixture of Figure 6.

g. Input power requirements. The connector assemblies shall perform satisfactorily under all applicable conditions specified herein with input voltages up to 500 ± 5 VDC.

4.6 Inspection methods.

4.6.1 Inspection, connector assembly. The connector assembly shall be inspected as follows.

4.6.1.1 Individual inspections. The connector assembly shall be inspected to ensure that it has been properly assembled and adjusted, that the electrical connections are secure and that the workmanship is as specified herein (see 3.6).

4.6.1.2 First article and quality conformance inspections.

a. The connector assembly shall be thoroughly examined to determine conformance to all the requirements specified herein (see 3.5.1). This inspection shall include a fit test within the parent bomb rack. See Table IV.

b. The connector assembly shall be examined to ascertain that the packaging conforms to contract requirements (see 5).

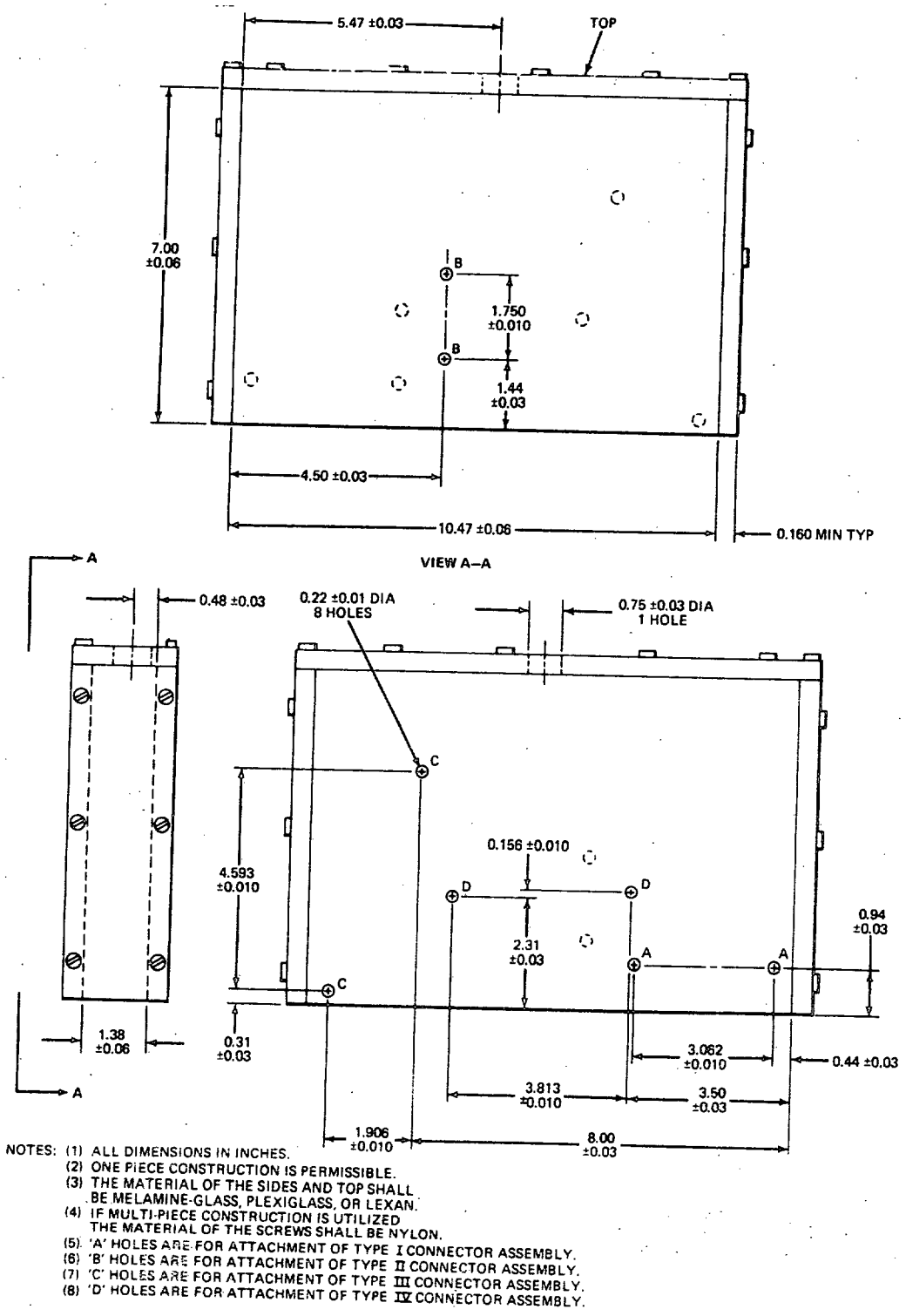
4.6.2 Functional inspections.

4.6.2.1 Connector pull-out force test, 0°. The MK-39 receptacle of the connector assembly shall allow the pull test fixture to disengage at 28 ± 12 pounds (12.7 ± 5.44 kilograms). The pull test fixture of Figure 6 shall be utilized. The support points of the connector assembly shall be secured to withstand a force normal to the lower jaw face of the MK-39 receptacle (tolerance of direction of pull: ±2 degrees). The test force shall be applied slowly and evenly. Damage to the connector assembly shall be cause for rejection.

4.6.2.2 Connector pull-out force test, 15°. The connector assembly shall be tested as specified in 4.6.2.1 except the test force shall be applied as shown in Figure 7. The disengagement force shall not exceed 60 pounds (27.2 kilograms). Damage to the connector assembly shall be cause for rejection.

4.6.2.3 Dielectric withstanding voltage test. The connector assembly shall be subjected to the dielectric withstanding voltage test in accordance with MIL-STD-202, Method 301 and the following:

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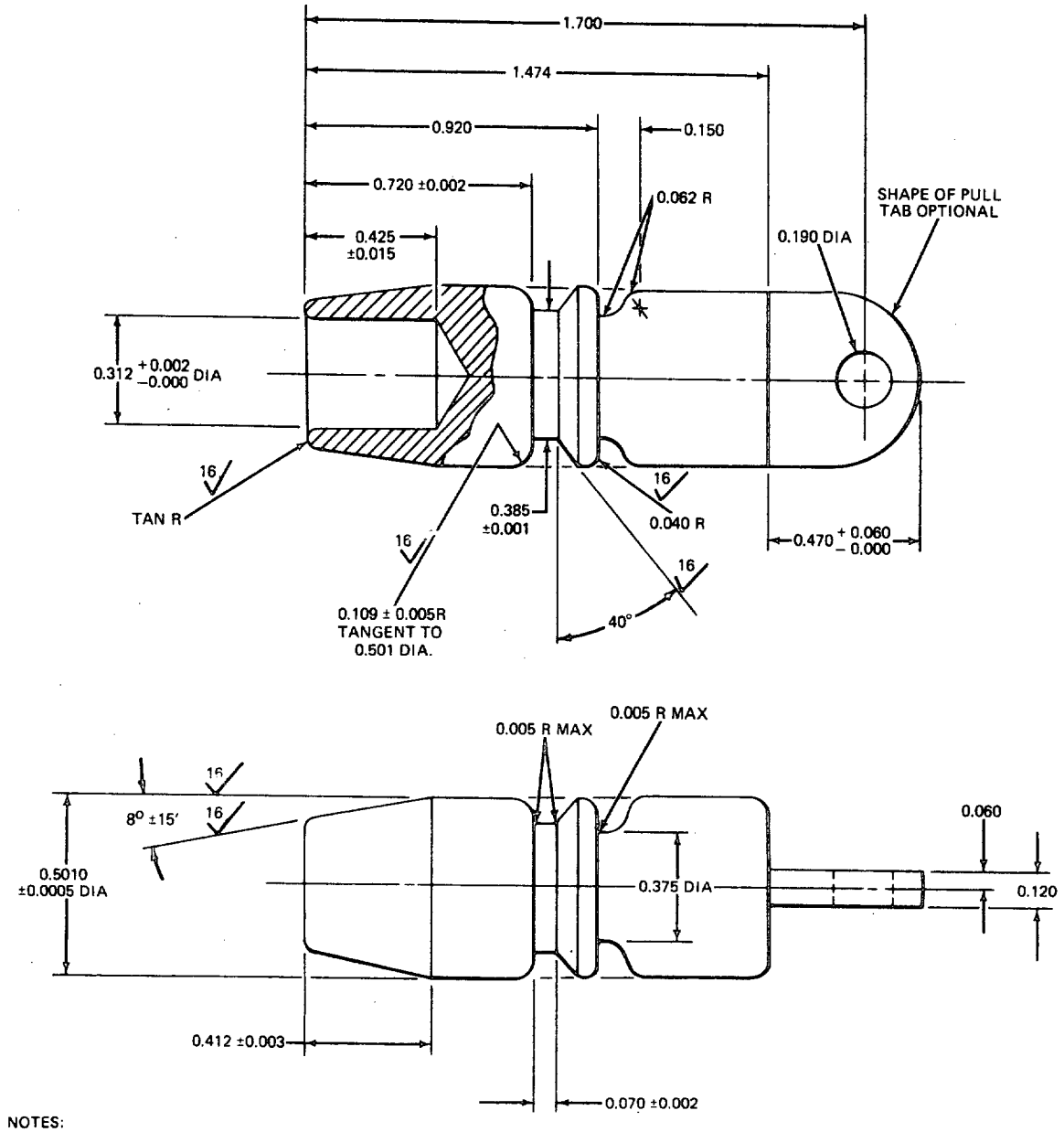


NOTES: (1) ALL DIMENSIONS IN INCHES.
(2) ONE PIECE CONSTRUCTION IS PERMISSIBLE.
(3) THE MATERIAL OF THE SIDES AND TOP SHALL BE MELAMINE-GLASS, PLEXIGLASS, OR LEXAN.
(4) IF MULTI-PIECE CONSTRUCTION IS UTILIZED THE MATERIAL OF THE SCREWS SHALL BE NYLON.
(5) 'A' HOLES ARE FOR ATTACHMENT OF TYPE I CONNECTOR ASSEMBLY.
(6) 'B' HOLES ARE FOR ATTACHMENT OF TYPE II CONNECTOR ASSEMBLY.
(7) 'C' HOLES ARE FOR ATTACHMENT OF TYPE III CONNECTOR ASSEMBLY.
(8) 'D' HOLES ARE FOR ATTACHMENT OF TYPE IV CONNECTOR ASSEMBLY.

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Figure 5. Salt Spray and Humidity Test Fixture

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NOTES:

1. ALL DIMENSIONS IN INCHES.
2. ALL DIAMETERS SHALL BE CONCENTRIC WITH THE 0.501 DIAMETER WITHIN 0.003 TIR.
3. MATERIAL: TOOL STEEL, TYPE T1.
4. HARDNESS: HRC 58-60.

TOLERANCES UNLESS OTHERWISE SPECIFIED:
DECIMALS: ±0.010
ANGLES: ±30'

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Figure 6. Pull Test Fixture

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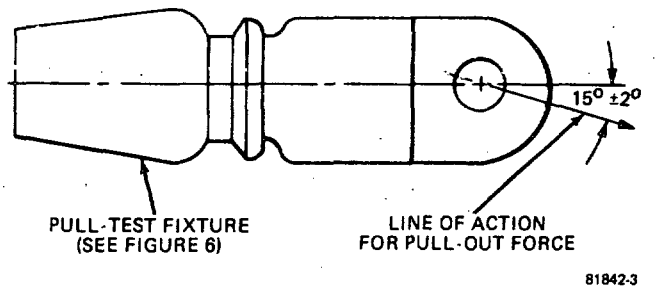


Figure 7. Application of Pull-Out Force (15° Attitude)

a. All actuator switches shall be actuated.

b. A potential of 1000 VDC shall be applied between the center conductor and shield of mating cable number 1 (see Figures 1 through 4) for 10 seconds minimum. Leakage current shall not exceed 0.5 milliampere.

4.6.2.4 **Continuity resistance test.** The connector assembly shall be subjected to the continuity resistance test in accordance with MIL-STD-202, Method 307 utilizing a test current of 0.5 (-0, +0.15) amperes. The continuity resistance shall not exceed 0.3 ohm when measured between center conductor to center conductor and shield to shield between the following cables with the applicable actuator switch(es) closed (see Figures 1 through 4). For this test, the overall length of the mating cables shall be 5 inches (12.70 centimeters) maximum.

a. Type I.

1. Mating cables, numbers 1 and 2.

2. Mating cables, numbers 1 and 3.

b. Type II.

1. Mating cables, numbers 1 and 2.

2. Mating cables, numbers 1 and 3.

c. Type III.

1. Mating cables, numbers 1 and 2.

2. Mating cables, numbers 1 and 3.

d. Type IV. Mating Cables, numbers 1 and 2.

4.6.2.5 **Insulation resistance test.** The connector assembly shall be subjected to 500 ± 50 VDC applied between conductors and ground. In addition, the connector assembly shall be subjected to the following tests.

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4.6.2.5.1 Type I. With actuator switch S1 closed and actuator switch S2 open (see Figure 1), the resistance between the center conductors of mating cable number 1 and mating cable number 2 shall be a minimum of 40 megohms with 500 ± 50 VDC applied. The test shall be repeated with actuator switch S1 open and actuator switch S2 closed.

4.6.2.5.2 Type II. With actuator switch S1 open (see Figure 2), the resistance between the center conductors of mating cable number 1 and mating cable number 3 shall be a minimum of 40 megohms with 500 ± 50 VDC applied.

4.6.2.5.3 Type III. With actuator switch S1 open (see Figure 3), the resistance between the center conductors of mating cable number 1 and mating cable number 2 shall be a minimum of 40 megohms with 500 ± 50 VDC applied.

4.6.2.5.4 Type IV. With actuator switch S1 open (see Figure 4), the resistance between the center conductors of mating cable number 1 and mating cable number 2 shall be a minimum of 40 megohms with 500 ± 50 VDC applied.

4.6.3 Environmental inspections. The connector assemblies shall be subjected to the following environmental inspections. Unless otherwise specified, the connector assemblies shall be installed in an applicable test fixture which will allow switch actuation and withstand the applied loads. Upon completion of each environmental inspection, the connector assemblies shall be visually inspected for damage due to exposure to environmental conditions. The connector assemblies shall be in accordance with the drawings listed in the data list for the connector assembly being procured.

4.6.3.1 High temperature. The connector assembly shall be subjected to a temperature of 70°C (158°F) and a relative humidity of less than 15 percent, for 2 hours after stabilization is attained. While still at temperature, the connector assembly shall be tested in accordance with 4.6.2.3, 4.6.2.4 and 4.6.2.5 as required. Within 1 1/2 minutes after completion of testing the connector assembly shall be tested in accordance with 4.6.2.1 and 4.6.2.2.

4.6.3.2 Low temperature. The connector assembly shall be subjected to a temperature of -56.7°C (-70°F) for 2 hours after stabilization is attained. While still at temperature, the connector assembly shall be tested in accordance with 4.6.2.3, 4.6.2.4 and 4.6.2.5 as required. Within 1 1/2 minutes after completion of testing, the connector assembly shall be tested in accordance with 4.6.2.1 and 4.6.2.2.

4.6.3.3 Humidity. The connector assembly shall be subjected to the humidity test of MIL-STD-810, Method 507, Procedure I. The actuator switch(es) shall not be actuated during the test and all mating cables except cable number 1 shall be attached. At the completion of testing, the connector assembly shall be tested in accordance with 4.6.2.3, 4.6.2.4 and 4.6.2.5. The connector assembly shall then be allowed to dry for a minimum period of 72 hours after which it shall be tested in accordance with 4.6.2.1 and 4.6.2.2.

4.6.3.4 Sand and dust. The connector assembly shall be subjected to the following sand and dust tests. The applicable bomb rack shall be utilized when testing in accordance with 4.6.3.4.3. Otherwise, testing shall be as specified in 4.6.3. For tests of 4.6.3, Type IV connector assemblies shall be used with a MER/TER ejector unit and the ejection shall be in a vertical down direction.

4.6.3.4.1 First dusting. The connector assembly shall be subjected to the dust test in accordance with MIL-STD-810, Method 510, Procedure I, steps 1 and 4 only. Type IV connector assemblies shall be installed in a centerline ejector unit. The actuator switch(es) shall not be actuated during the test and all

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mating cables except cable number 1 shall be attached (see Figures 1 through 4). At the completion of testing, the connector assembly shall be tested in accordance with 4.6.2.1 and 4.6.2.2. The test of 4.6.2.1 shall be repeated a total of three times prior to tests of 4.6.2.2.

4.6.3.4.2 Second dusting. The connector assembly, with the pull test fixture (see Figure 6) installed, shall be resanded in accordance with 4.6.3.4.1 and the tests of 4.6.3.4.1 shall be repeated.

4.6.3.4.3 Frozen sand. The connector assembly shall be sanded in accordance with 4.6.3.4.1 except at the completion of testing, the unit shall be subjected to the icing test of 4.6.3.7 with the exception of step b. At the completion of testing, while at icing conditions, the item shall be tested in accordance with 4.6.2.2.

4.6.3.5 Salt fog. The connector assembly shall be subjected to the salt fog test of MIL-STD-810, Method 509, Procedure I. Parts of the connector assembly normally protected when installed on the aircraft shall be adequately protected during the test (see 4.5). The actuator switch(es) shall not be actuated during the test and all mating cables except cable number 1 (see Figures 1 through 4) shall be attached. The connector assembly shall be allowed to dry for a minimum period of 24 hours after which it shall be tested in accordance with 4.6.2.3, 4.6.2.4 and 4.6.2.5. At the completion of 72 hours of drying, the connector assembly shall be tested in accordance with 4.6.2.1 and 4.6.2.2.

4.6.3.6 Vibration.

4.6.3.6.1 General. The applicable vibration tests specified herein shall be performed in each of three mutually perpendicular axes in any order. All testing shall be completed in one axis prior to testing in the next axis. At the conclusion of each test or significant portion thereof, the item shall function as described herein. The test fixture of 4.5f allows up to four connector assemblies to be tested together. Chatter or open circuits of the connector assemblies in excess of the tolerances in 4.6.3.6.2, during vibration from any cause shall be considered a failure. Structural damage to the connector assemblies shall be considered a failure. The procuring activity shall decide when structural damage, if any, is to be considered significant and cause for rejection.

4.6.3.6.2 Test item operation. The connector assembly shall be subjected to the following functional checks during vibration:

a. The connector assembly shall be monitored during vibration in each axis to determine if the circuitry between the center pins of the MK-39 receptacle and the input electrical connector of the connector assembly is open for more than a total of 3 ± 0.3 milliseconds during each 20 ± 0.2 millisecond period of vibration when 500 VDC is applied. The voltage shall be applied between the center conductor and the shielding of the mating cable of the input connector (see Figures 1 through 4). The circuit shall be considered open when the output voltage is less than 480 volts at mating cable number 1. The existence of circuit openings exceeding 3 ± 0.3 milliseconds shall constitute a failure. The Switch Bounce Test Set (SA-A-20400), or equivalent, shall be used to determine whether the connector assembly meets the above requirement. The procuring activity shall furnish the test set to the government approved testing laboratory for use during first article and quality conformance testing.

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b. The connector assembly shall be subjected to a 500 VDC test during each axis of vibration, with the actuator switch open or closed. The connector assembly shall withstand ten, 500 ± 50 VDC pulses of 5 ± 1 second duration, spaced 5 ± 1 second apart. There shall be no indications of a dead short (leakage current shall be less than 0.5 milliampere).

c. At the completion of each axis, the connector assembly shall be tested in accordance with 4.6.2.1, 4.6.2.2, 4.6.2.3, 4.6.2.4 and 4.6.2.5.

4.6.3.6.3 Mounting and control techniques. The connector assemblies shall be attached to the vibration exciter table by means of the fixture described in 4.5f. Vibration amplitudes and frequencies shall be measured by techniques that will not significantly affect test item input control or response. The transverse motion at the input monitoring point shall be minimized, and should be limited to 100 percent of the input motion. The vibration control shall be mounted on the top of the fixture cube (refer to DL 1011AS100).

4.6.3.6.4 Test. The connector assembly shall be vibrated in accordance with the applicable vibration curve of Figures 8 through 11. Vibration duration shall consist of 1 hour in each axis. The following conditions and tests shall be applied to the connector assembly during each axis.

a. For the first 30 minutes of vibration the actuator switch shall be closed.

1. During this time, the connector assembly shall be monitored for chatter, in accordance with 4.6.3.6.2a for the first 15 minutes.

2. During the remaining 15 minutes of vibration, the test of 4.6.3.6.2b shall be conducted.

b. For the last 30 minutes of vibration the actuator switch shall be open.

1. During this time, the connector assembly shall be monitored for chatter, in accordance with 4.6.3.6.2a for the first 15 minutes.

2. During the remaining 15 minutes of vibration, the test of 4.6.3.6.2b shall be conducted. In addition, the test of 4.6.2.1 shall be conducted. Only one force measurement shall be made and the pull test fixture shall not be reinstalled after the measurement.

c. At the completion of each vibration axis, the test of 4.6.3.6.2c shall be conducted.

4.6.3.7 Icing. The connector assemblies shall be subjected to the icing test as follows:

a. The connector assembly shall be installed in its applicable parent bomb rack (see Table IV). Type IV connector assemblies shall be used with a MER/TER ejector unit and the ejection shall be in a vertical down direction. A test store simulating the applicable test store (see Table V) with a MK-122 switch attached, shall be loaded onto the bomb rack and the connector of the MK-122 switch connected to the MK-39 connector of the connector assembly. For Type I connector assemblies, the actuator switch for the 30-inch (76.20-centimeter) suspension hooks of the parent bomb rack shall be closed during this test.

b. The test store shall be ejected at room ambient conditions, using the cartridges specified in Table V. During the ejection the following time intervals shall be measured.

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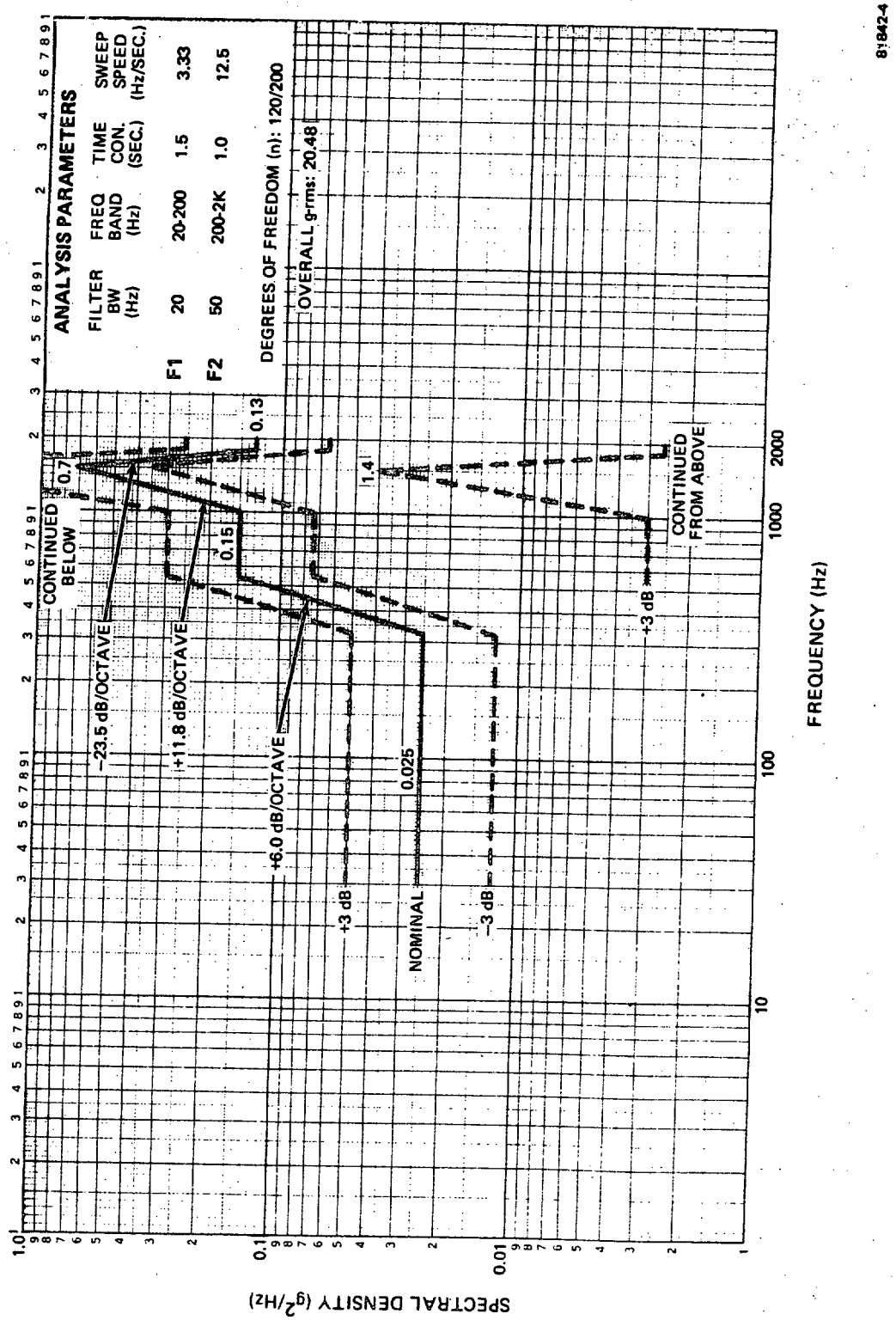


Figure 8. Vibration Curve for Type I Connector Assemblies

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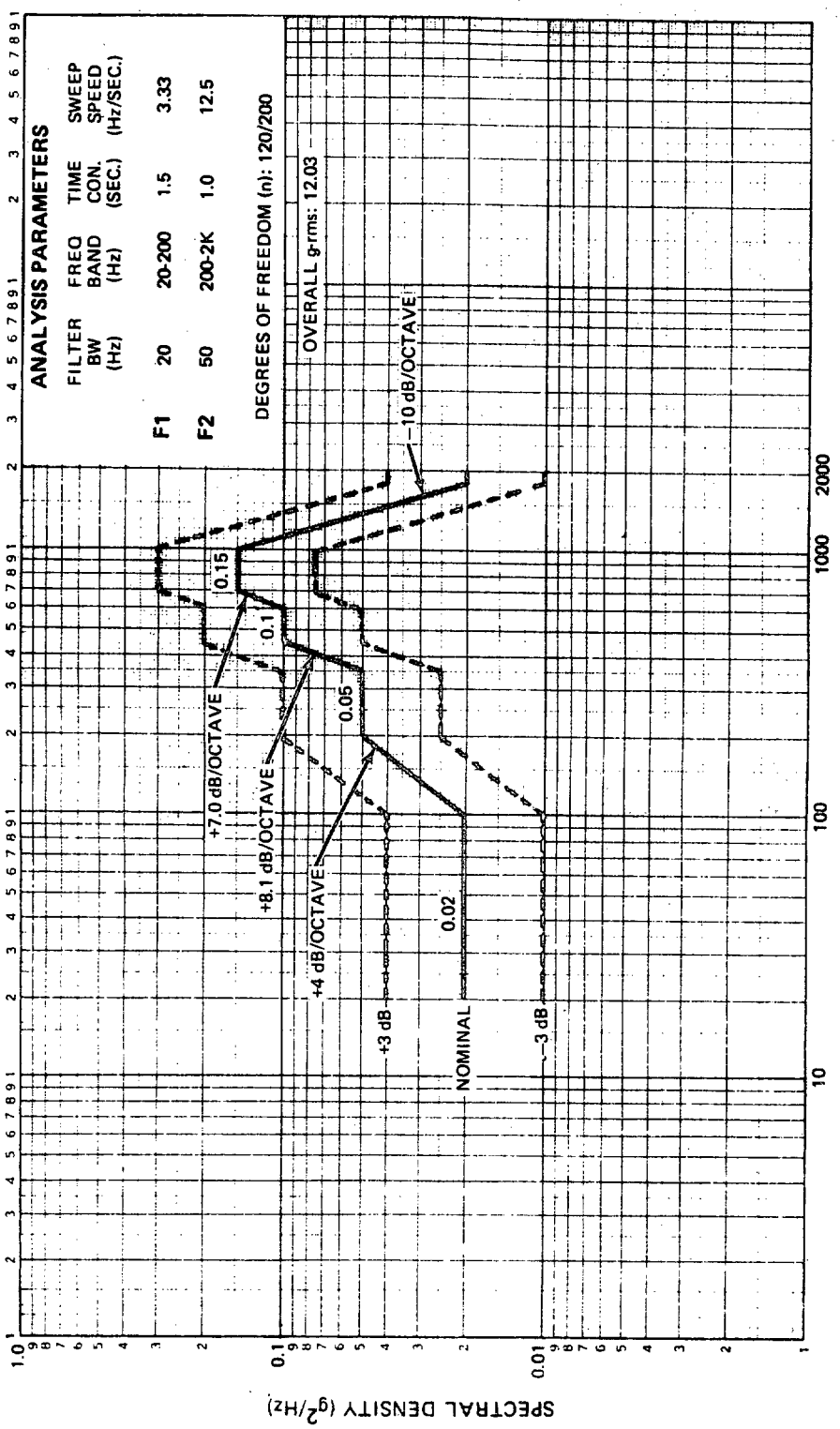


Figure 9. Vibration Curve for Type II Connector Assemblies

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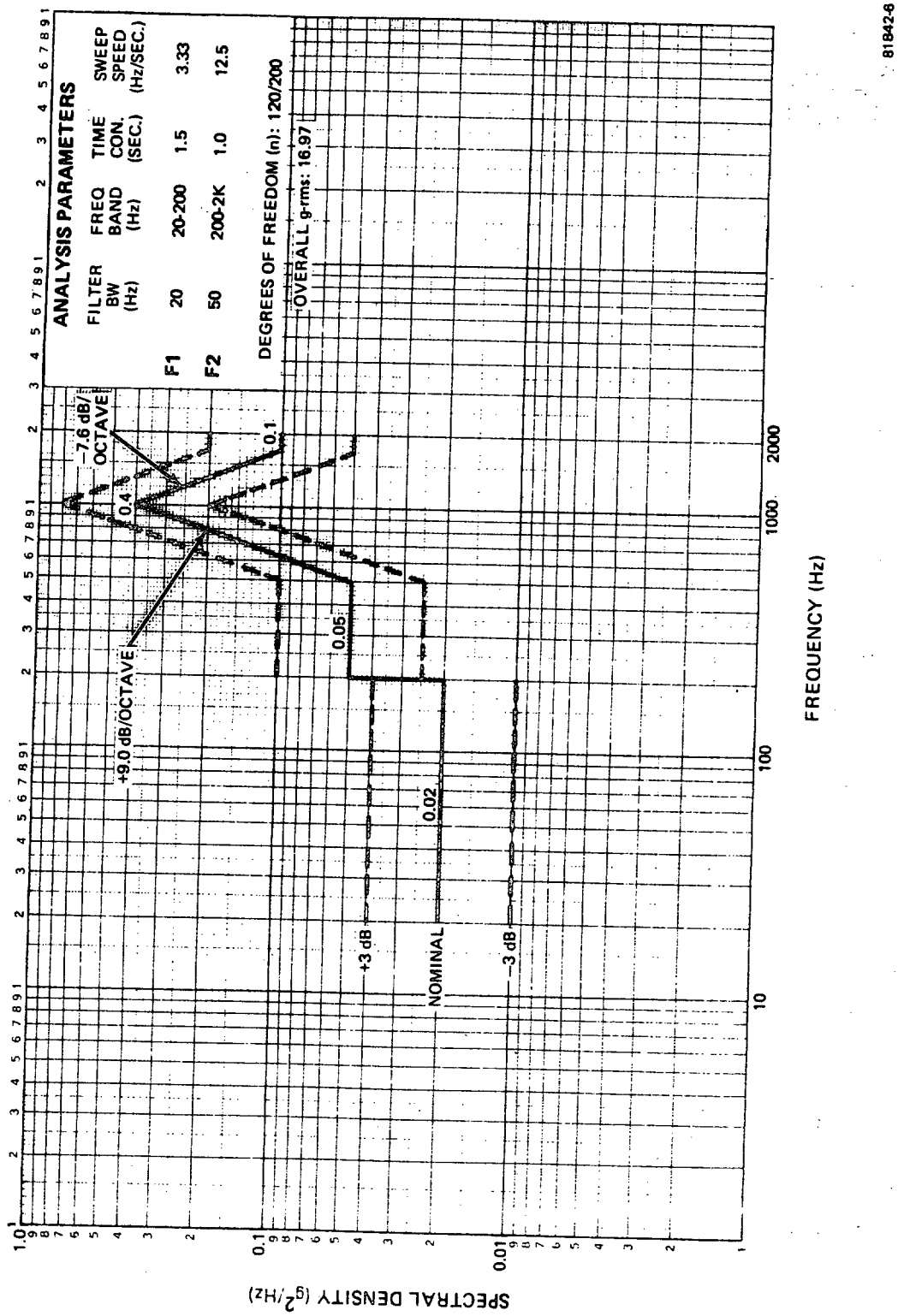


Figure 10. Vibration Curve for Type III Connector Assemblies

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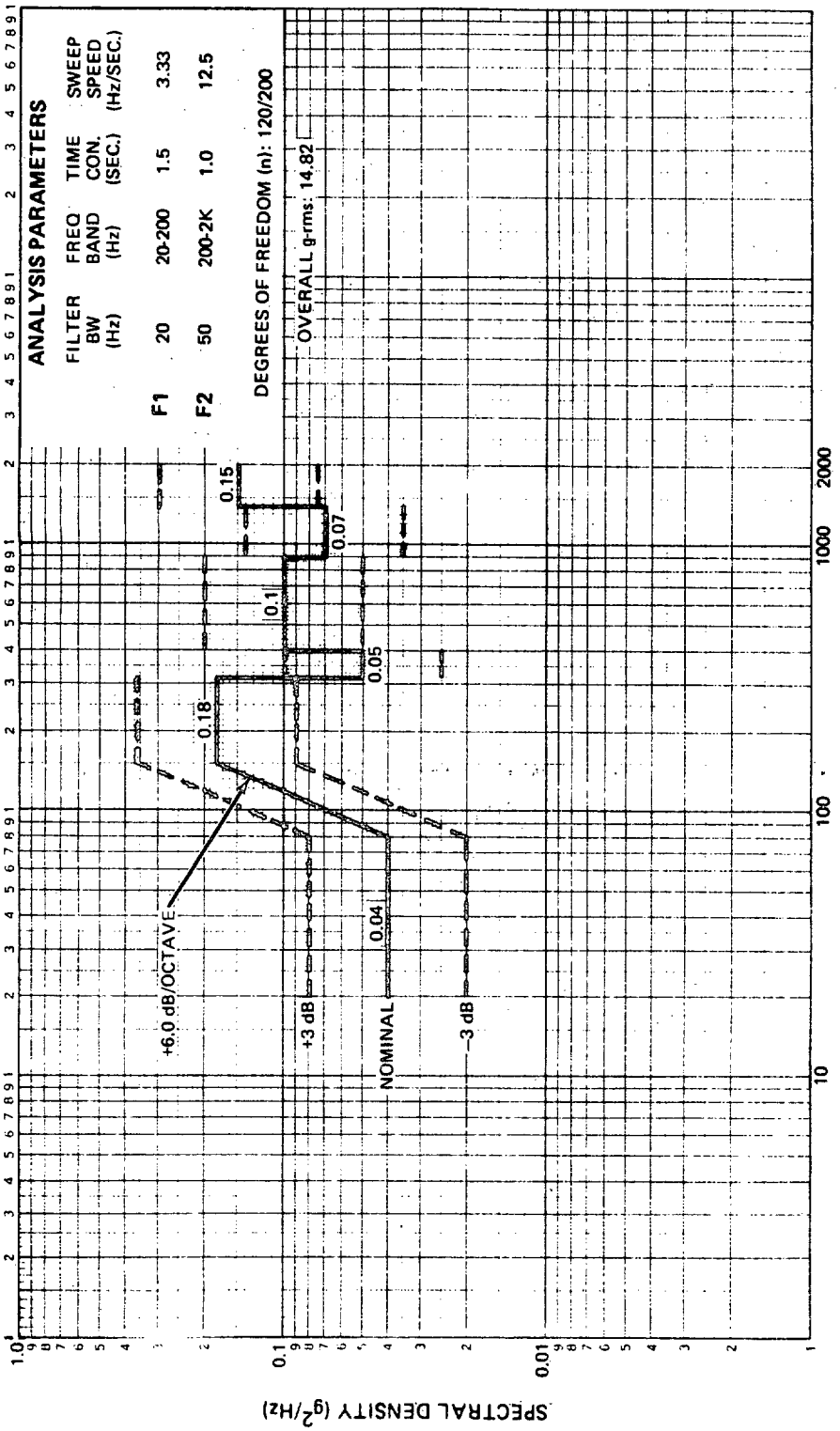


Figure 11. Vibration Curve for Type IV Connector Assemblies

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1. Fire pulse to rack actuator switch initial closure.
2. Fire pulse to rack actuator switch final closure.
3. Fire pulse to MK-122 switch closure.
4. Fire pulse to MK-122 switch disconnect.
5. Interlock switch/MK-122 switch overlap (fuze charging period).

The following conditions shall apply during the instrumented ejection:

Types I, II and IV. The time interval between final closure of the actuator switch and loss of continuity between the center pins of the MK-39 receptacle and the electrical connector of the MK-122 switch (fuze charging period) shall be 30 milliseconds minimum for Types I and IV connector assemblies and 17 milliseconds minimum for Type II connector assemblies.

Type III. The total time the actuator switch remains closed shall be a minimum of 70 percent of the time between initial closure of the MK-122 switch contacts and loss of continuity between the center pins of the MK-39 receptacle and the electrical connector of the MK-122 switch, but shall not be less than 10 milliseconds.

c. The connector assembly, prepared as specified in a. above, shall be subjected to a temperature of -56.7°C (-70°F), after stabilization is attained, for 1 hour. The connector assembly shall then be transferred to an ambient condition of 37.8°C (100°F) and a relative humidity of 90 percent until all evidence of frost has disappeared. Retaining as much moisture as is possible, the connector assembly shall be transferred to an ambient condition of -56.7°C (-70°F) for 30 minutes. Within 1 1/2 minutes after completion of the test, the test store shall be ejected as specified in a. above. Data shall be obtained and meet the requirements as specified in b. above.

TABLE V

CONNECTOR ASSEMBLY, STORE AND EJECTION CARTRIDGE RELATIONSHIP

CONNECTOR ASSEMBLY	STORE	EJECTION CARTRIDGE
Type I	MK-81 250 Pounds (113.4 Kilograms)	Two MK-2 or Two MK-125
Type II	MK-81 250 Pounds (113.4 Kilograms)	Two MK-2 or Two MK-125
Type III	MK-81 250 Pounds (113.4 Kilograms)	Two MK-2 or Two MK-125
Type IV	MK-81 250 Pounds (113.4 Kilograms)	One MK-2

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4.6.4 Life test. The connector assembly shall be subjected to a life test consisting of 500 engagements and disengagements of the pull test fixture of Figure 6 from the MK-39 receptacle. The support points of the connector assembly shall be secured to withstand a force normal to the face surface of the mating connector. During every 50th disengagement, the pull-out force shall be measured as specified in 4.6.2.1. Failure to meet the requirements of 4.6.2.1 or evidence of structural damage to the equipment shall constitute a failure and shall be cause for rejection.

5. PACKAGING

5.1 Connector assembly preservation and packaging. Preservation and packaging shall be in accordance with Level A, B or C of MIL-STD-794, and as defined in the contract, without the use of preservation compound.

5.2 Packing. Shipping containers shall contain one each packaged connector assembly. The container shall be suitable for Level A, B or C as specified in MIL-STD-794.

5.3 Unitized loading. Unitized loading for the connector assemblies is authorized as specified in MIL-STD-794.

5.4 Marking. All shipping containers shall be marked in accordance with the requirements of MIL-STD-129 with the following information:

CONNECTOR ASSEMBLY, ELECTRIC FUZING PARENT BOMB RACK
MILITARY SPECIFICATION NO. MIL-C-81842(AS)
PRESERVED (DATE)
DOMESTIC OR EXPORT PACKED (AS APPLICABLE)
GOVERNMENT ORDER NO. (OR CONTRACT NO. IF ORDER NO. IS NOT ASSIGNED)
NAME OF SUPPLIER (AND NAME OF MANUFACTURER IF NOT THE SAME)
MANUFACTURER'S CODE IDENTIFICATION NUMBER

6. NOTES

6.1 Intended use. The connector assemblies are intended for use as a part of an aircraft electrical bomb fuzing system to transfer an arming signal from the aircraft fuze function control set to the stores attached to the bomb racks.

6.2 Ordering data. Procurement documents shall specify the following.

6.2.1 Procurement requirements.

a. Title, number and date of this specification.

b. Selection of applicable levels of preservation, packaging and packing required (see 5.1 and 5.2).

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c. Samples subjected to sampling inspection shall not be considered or accepted as part of the contract or purchase order (see 4.4.2).

d. Sampling inspection selection (see 4.4.2).

e. Number of samples and inspection sequence for production lots in excess of 300 units (see Table III).

f. Applicable drawing package for type of connector assembly required (see 3.5.1).

g. Name and location of government representative reponsible for random selection of inspection samples (see 4.4.2).

h. Items of data required (see 6.3).

i. Name and location of government approved test laboratory.

6.2.2 Contract provision. Contracts shall specify the following provision for first article inspection.

6.2.2.1 First article. When a first article is required for inspection and approval (see 3.1, 4.3, 6.2 and 6.3), the contract or purchase order shall specify the following provision for first article inspection. When a contractor is in continuous production of bomb racks from contract to contract, consideration should be given to waive the first article inspections. If inspection is required, indicate:

a. If first article inspections are conducted at the contractor's plant or a government approved laboratory, an inspection report shall be forwarded to the procuring activity for verification.

b. That the approval of first article samples or the waiving of the first article inspection shall not relieve the contractor of his obligation to fulfill all other requirements of the specification and contract.

6.3 Contract data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of 7-104.9(n) of the Armed Services Procurement Regulations, the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of ASPR-7-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data Requirement	Applicable DID
4.3	First Article Inspection Reports	DI-T-5329 - Inspection Test Reports
4.4	Quality Conformance Inspection Reports	DI-T-5329 - Inspection Test Reports

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(Copies of data item descriptions required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

6.4 **Quality assurance definitions.** Definitions for quality assurance terms are in accordance with MIL-STD-109.

Preparing Activity
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE
MIL-C-81842A(AS) CONNECTOR ASSEMBLIES FOR BOMB RACK ELECTRIC FUZING PROVISIONS

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO (If "Yes", in what way?)

4. REMARKS

SUBMITTED BY (Printed or typed name and address - Optional)

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